

Arthur Lau

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Profile Summary

A **Results-driven** Senior Software Engineer in **full-stack** development specializing in **React(Typescript), Next.js, Python, C#(.Net), Flutter, Cloud Service. Adaptable** in delivering **scalable, secure, and robust** solutions across diverse domains such as vending automation, ERP, IoT, finance and gaming with proficiency in designing **system architecture**, implementing **performance optimization** and driving **project accountability**. Strong in **cross-functional collaboration** with PMs, analysts and engineers to yield business-aligned outcomes. Consistently pursue **technical craftsmanship** through team mentorship, and maintainable code practices.

Experience

Senior Software Engineer, IVM Tech Limited

Aug 2023 – Present

A leading vending machine company in the regional market, serving 2k+ sellers and 500k+ customers.

- Architected and led an ERP digital transformation by **.NET, MQTT, Redis, React and Svelte(TailwindCSS)**, reducing manual operation cost by 50%.
- Spearheaded a 0-to-1 development of **distributed** IoT platform reaching 250K+ users using **Next.js** with **Turborepo** and **ShadCN, Flutter**, and **MSSQL**.
- Orchestrated **high-availability** mobile payment Infrastructure and hardware Integration using **Java, Kotlin** for 1,000+ retail clients (Octopus, AliPay).
- Designed a high-throughput **telemetry and monitoring dashboard** using **NestJS** and **MSSQL**, optimizing query execution across 1,000+ IoT nodes.
- Automated **CI/CD** on **AWS** and **Azure** with comprehensive testing using **Vitest, MSTest** and **Playwright**, minimizing deployment duration by 60%.
- Designed **cross-functional technical** specifications across teams with 10+ smart fridge partners ensuring consistent interoperability.

Software Engineer, Makeitfun Studio

Jul 2022 – Aug 2023

A digital creative studio providing fun experiences via various gamification forms with an average of 100k+ active users.

- Constructed **high-traffic SEO-optimized** web experiences with **React + Next.js(Mui)** serving 500k+ users, boosting reach by 45%.
- Designed scalable db and logic with **RESTful** and **GraphQL** for 10+ projects by **Express.js(Sequelize), MongoDB, PostgreSQL, Docker** and **Kubernetes**.
- Delivered a 3D real-time metaverse voting system from **Mozilla Hubs** using **React, Three.js, Elixir, WebRTC** to enable low-latency interaction.
- Programmed a Cross-Platform **AR Flutter** app for a Disney Studio using **MediaPipe** and **Vuforia** in **Unity** and **C#** with 70k+ served customers.

Software Engineer, Flexsystem Limited

Jun 2021 – Jul 2022

An international enterprise SaaS provider serving 2000+ clients in 30+ countries.

- Tailor-made a **SaaS ERP** system for International client with over 2M+ users using **React, C#** with providing on call support.
- Optimized QA workflows by implementing **RPA(UIPath)** for **stress** and **unit** tests with **Jest** and **Selenium**, cutting execution time by 50%.

Software Engineer, Excohk Limited

Oct 2020 – Jun 2021

A Fintech-focused cultural media startup that bridges art, storytelling, and creative branding for 50+ institutions and communities.

- Developed a **Secure OAuth2 Booking & Identity System** from **React Native, Expo, Spring Boot** for booking 50+ large-scale cultural events.
- Integrated a **temporal storage solution** via **DynamoDB** and **Firebase**, improving responsiveness and visitor interaction reliability.

Early Technical Experience

Jun 2018 – Sep 2020

- **East Technology(2020)**: Implemented UX on SPA across 20+ brands by **jQuery, JavaScript**, and a lightweight **CMS** using **Apache** and **phpMyAdmin**
- **Yungfeng Financial Group(2019)**: Programmed **Flask** Server API of Youyu Smart Trade App towards **MySQL**, ensuring production level reliability.
- **Freelance(2018-2019)**: Programmed SPA website for local business and self-driven projects

Projects

Vigilant Stream - Developed a distributed, risk ingestion mock engine to process high-velocity security signals by **Python** and **Elixir**

Prompt Pounder - Engineered a **Manifest V3** extension to solve the context-switching productivity gap caused by **LLM** inference delays.

Arkham Analysis - Built an analytic engine for card games using **FastAPI, PostgreSQL, and Angular** to model and visualize 1k+ data.

Reals - Created an agentic LLM **Flutter** app for custom dating itineraries with **OpenAI** and **FastAPI**, providing click-to-plan experience.

Mozilla Hubs - Integrated open source 3D chess with Avatar Maker and observer mode using **React, Three.js, Elixir, WebRTC, Express.js**

Automatic Chord Identification - Coded A Symbolic AI machine learning musicology system for chord identification using **Python**

Skills

Programming Languages: **C# (.NET Core), Java, Python, TypeScript/JavaScript, Flutter, Go, Kotlin, Elixir**.

DevOps & Cloud Platforms: **Git, Docker, Kubernetes, Jenkins, Azure, AWS, Cloudflare, GCP, Terraform**.

Programming Practise: **Design Pattern, Architecture(MVC, microservice, Domain Driven Design etc), SOLID, version control, Agile**

Database & Messaging: **PostgreSQL, MSSQL, SQLite, RabbitMQ, MQTT, MongoDB, Redis, MySQL, GraphQL, DynamoDB, Firebase**

Education

Chinese University of Hong Kong – Bachelor's Degree in Computer Science

Jun 2021